



Particle swarm optimizer with crossover operation

Yonggang Chen^{a,b}, Lixiang Li^{c,*}, Jinghua Xiao^d, Yixian Yang^c, Jun Liang^c, Tao Li^e

^a Information Engineering College, Henan University of Science and Technology, Luoyang 471023, China

^b School of Science, Beijing University of Posts and Telecommunications, Beijing 100876, China

^c Information Security Center, State Key Laboratory of Networking and Switching Technology, Beijing University of Posts and Telecommunications, Beijing 100876, China

^d State Key Laboratory of Information Photonics and Optical Communications, School of Science, Beijing University of Posts and Telecommunications, Beijing 100876, China

^e College of Computer Science, Sichuan University, Sichuan 610065, China

ARTICLE INFO

Keywords:

Swarm intelligence
Particle swarm optimization
Crossover operation
Global optimization

ABSTRACT

A particle swarm optimization algorithm with crossover operation (PSOCO) is proposed. In the proposed PSOCO, two different crossover operations are employed in order to breed promising exemplars. By performing crossover on the personal historical best position of each particle, the effective guiding exemplars are constructed and they maintain a good diversity. In turn, these high quality exemplars are used to guide the evolution of particles. PSOCO is two-layer particle swarm optimization with positive feedback mechanism. In order to test the performance of PSOCO, we use a set of widely used benchmark functions. The experimental results demonstrate that the proposed PSOCO is a competitive optimizer in terms of both solution quality and efficiency.

1. Introduction

Lots of control and decision problems can be transformed into optimization problems (Chang, 2009; Chang and Shih, 2010; Pan et al., 2008; Jarboui et al., 2008). The optimization method and its application in the research of complex system is a challenging subject. With the development of technology, many problems encountered in engineering practice are becoming more and more complex. Some traditional computing methods are used to solve these problems. However, these methods often have the problems of high computational complexity and require optimization problems to be represented by continuous and differentiable function. Due to the limitations of traditional algorithms, lots of metaheuristic algorithm are designed to find satisfactory and approximate solutions for hard optimization problems.

In the research of nature-inspired metaheuristic optimization algorithms, the research of swarm intelligence algorithms is one of the most active fields. Some of the well known swarm intelligence algorithms are developed during the last two decades. Popular swarm intelligence optimizers include particle swarm optimization (PSO) (Eberhart and Kennedy, 1995; Kennedy and Eberhart, 1995), ant colony optimization (Yao et al., 2015), chaotic ant swarm algorithm (Li et al., 2006), biogeography-based optimization (Simon, 2008), etc. Many practices and experiments have shown that swarm intelligence algorithms are

more effective and efficient in solving hard optimization problems (Zhang et al., 2010; Peng et al., 2013; Cai et al., 2012).

As a population-based and global optimizer, PSO was initially proposed by James Kennedy and Russell Eberhart in 1995. PSO was inspired by emulating the foraging behaviors of birds and fish schooling. Compared with other swarm intelligence algorithms, PSO is easy to be implemented and it has a good performance in solving many real world problems (Zhang et al., 2015). Owing to its simple concept, PSO has been the most popular and well-known algorithm. So far, PSO has been successfully applied in solving lots of scientific and engineering problems (Lu et al., 2015; Shen et al., 2014; Lin et al., 2016; Haddar et al., 2016; Gaxiola et al., 2016). However, lots of experiments have shown that traditional PSO algorithms cannot guarantee to find the global optimal solution and are easy to fall into local optima. In order to improve the performance of PSO, lots of PSO variants have been proposed. These variants can be mainly classified into the following two categories.

The first method is based on self-change strategy. Shi and Eberhart proposed that the inertia weight with decreased linearly was a good choice in 1998 (Shi and Eberhart, 1998). The inertia weight with fuzzy control has been proposed further (Shi and Eberhart, 2001). Other control strategies for parameters embedded in the particle motion equation have also been proposed (Taherkhani and Safabakhsh, 2016; Guo et al., 2006; Ratnaweera et al., 2004; Ardizzone et al., 2015; Olivas et

* Corresponding author.

E-mail address: lixiang@bupt.edu.cn (L. Li).

al., 2016). The main role of these additional parameters is to adjust the particle's flight. In PSO, it is a good idea that new topological structures are adopted. Some other topologies have been proposed (Kennedy, 1999). In 2004, Mendes et al. proposed a fully informed PSO (FIPS) (Mendes et al., 2004). In FIPS, the historical best positions of several neighbors were used to adjust the position of each particle. A new updating model of velocity was proposed in the comprehensive learning PSO (CLPSO) (Liang et al., 2006). In CLPSO, the velocity updating rule of each particle was based on different historical personal best positions.

The second method is to combine other strategies in PSO (Liu et al., 2015; Zhao et al., 2016; Wu et al., 2014; Li et al., 2015a; Tang and Fang, 2015; Samma et al., 2016; Zhang et al., 2014; Wang et al., 2016; Valdez et al., 2017). Based on orthogonal experimental design (OED), Zhan et al. proposed orthogonal learning PSO (OLPSO) (Zhan et al., 2011). Because OED can mine for useful information, an effective guidance exemplar can be constructed by orthogonal learning (OL) strategy. In this way, particle swarm can fly to promising position. The experiments showed that the OL strategy could give PSO more robust and reliable in numerical optimization problems. In ALC-PSO (Chen et al., 2013), Chen et al. introduced the aging mechanism into PSO. This mechanism was used to change the historically best position of the entire swarm. The simulation results showed that this method enhanced the optimization performance of PSO. Social learning mechanisms was introduced into PSO by Cheng and Jin (2015). In this version, other better particles based on the current swarm were used to guide each particle to fly. The experiment results showed this idea was more successful. Using multi-swarm technique is also a good method (Xu et al., 2015; Chen, 2011; Gülcü and Kodaz, 2015). Because of the information exchange among different groups, multi-swarm technology can better balance the exploiting and exploration ability of the algorithm. Liang and Suganthan designed a dynamic multi-swarm PSO (DMS-PSO) (Liang and Suganthan, 2005). In DMS-PSO, because these small swarms were regrouped continuously, the whole swarm maintained better diversity. A multi-layer PSO (MLPSO) was proposed by Wang et al. (2014). The MLPSO can avoid lapsing into the local optimum by increasing the number of the swarm layers. Multi-layer strategy can be regarded as a multi-swarm technique.

So far, lots of crossover operators are proposed. The crossover operator has always been regarded as one of the main search operators in genetic algorithm, because it exploits the available information in previous samples to influence future searches (García-Martínez et al., 2008). The crossover operator can combine parts of good solution to form new potential solution (Singh et al., 2011). Due to its advantages, crossover operator can improve the performance of PSO. In this paper, the scheme based on two crossover operators is designed. PSOCO is different from other PSO algorithms with crossover operator in many aspects. Differential Evolutionary (DE) crossover operator and the modified velocity update strategy are adopted in PSOCO. A dynamic adjustment strategy is adopted to enhance exploitation capability. The main loop of the PSOCO is composed of two layers. In the top layer, two crossover operators are used to construct promising exemplar. Particles in the bottom layer are guided by exemplar. The PSOCO establishes a positive feed back to accelerate the population to find the optimum. Moreover, the two-layer evolution mechanism of the PSOCO can accelerate the convergence speed of the population.

The main contributions of this paper can be summarized as follow:

(1) In this paper, two crossover operators and the velocity update scheme with learning model are incorporated into the proposed approach.

(2) The role of crossover operation is investigated. The influences of some parameters are investigated. According to some experimental results, a better parameter configuration is found.

(3) A dynamic strategy is proposed. This strategy is simple and can enhance the exploitation ability of the proposed approach. Our experimental results show that PSOCO has better optimization performance on some benchmark functions.

The remainder of this paper is organized as follows. Section 2 introduces the basic PSO and gives a review about the crossover-based PSO algorithms. Section 3 presents the PSOCO in detail. In Section 4, the experimental results are discussed and analyzed between several state-of-the-art PSO algorithms and PSOCO. Some analysis of PSOCO is further discussed. Conclusions and outlook of work are given in Section 5.

2. Related works

2.1. The basic PSO theory

In PSO, a particle is used to represent the potential solution of the optimization problem. The whole particle swarm flies in the search space to search for the global optimum. In a D-dimensional hyperspace, the following two vectors are related to the i th particle ($i = 1, 2, \dots, N$), where N denotes the population size. One is the velocity vector $V_i = (v_{i1}, v_{i2}, \dots, v_{id}, \dots, v_{iD})$, and the other is the position vector $X_i = (x_{i1}, x_{i2}, \dots, x_{id}, \dots, x_{iD})$, where $d \in [1, D]$. Firstly, the velocity and the position of each particle are initialized by random vectors with the corresponding ranges. Two equations are used to adjust the position. The position of the particle on each dimension is updated using the following equations:

$$v_{id}^{t+1} = wv_{id}^t + c_1 rand_1^t (P_{id}^t - x_{id}^t) + c_2 rand_2^t (P_{gd}^t - x_{id}^t) \quad (1)$$

$$x_{id}^{t+1} = x_{id}^t + v_{id}^{t+1} \quad (2)$$

where w is named as inertia weight. c_1 and c_2 are called the acceleration coefficients. $rand_1$ and $rand_2$ are two uniformly distributed random numbers in range $[0, 1]$. d denotes one dimension of a particle. P_g denotes the best position that the whole swarm has found so far. In addition, the flying velocity is limited to a reasonable range. A positive value V_{MAXd} is used to clamp the updated velocity. If $|v_{id}|$ exceeds V_{MAXd} , then the velocity is set to $\text{sign}(v_{id})V_{MAXd}$.

2.2. Crossover-based PSO algorithms

Crossover operator can enhance the information sharing between particles and prevent the premature convergence of swarm. In the past ten years, lots of meaningful work has been done in this field. For example, the work in Pant et al. (2007) introduced the quadratic crossover operator in PSO. This nonlinear crossover operator made use of the three particles to produce a better particle in the search space. Ref. Wang et al. (2008) presented PSO with a novel multi-parent crossover operator. According to multi-parent crossover operator equation, the offspring still stayed in the linear space by the four different parents. In order to achieve a successful recombination, a selection operator was employed. In Zhang et al. (2013), the crossover operator was implemented by borrowing the concept of linear combination of two vectors, where one vector was the swarm member and the other vector was selected randomly from elite set. In Tawhid and Ali (2016), arithmetical crossover operator and Nelder–Mead method are used to solve global optimization problems. In Duong et al. (2010), a hybrid of PSO and a GA (HEA) is proposed. In HEA, the BLX-crossover is used to produce offspring. In Chen (2012), particle swarm optimization with *pbest* crossover is proposed. The algorithm can achieve large improvements on Black-Box Optimization Benchmarking functions by crossing the personal best positions. Ref. Miao et al. (2009) presents Dynamic PSO with arithmetic crossover (DP-PSO). Crossover operator was effective for high-dimensional multimodel benchmarks in DP-PSO. Hao et al. (2007) proposed a PSO based on crossover, where the arithmetic crossover operator is adopted in one dimension of two different individual best position. Xie and Yang (2010) developed a novel crossover operator for particle swarm algorithm, where a Laplace crossover operator was employed to generate good candidate solutions. In Engelbrecht (2016), an extensive review of PSO algorithms that make use of some form

of crossover operator and an empirical analysis of the performance of these algorithms were provided. Furthermore, some crossover-based PSO algorithms have been successfully applied in solving other problems (Singh et al., 2011; Li et al., 2009; Park et al., 2007).

3. PSO algorithm with crossover operation (PSOCO)

In PSO, P_i and P_g determines the search behaviors of particles. These two vectors play an important role in PSO. According to the research results of CLPSO (Liang et al., 2006), this search model may make the particle oscillates in some cases or may make the particle trap into a local optimum in other cases, which causes premature convergence.

In order to solve the above problems, lots of variant PSO schemes have been developed. The following velocity update rule is very effective and widely adopted (Liang et al., 2006; Zhan et al., 2011; Li et al., 2015b).

$$v_{i,d}^{t+1} = wv_{i,d}^t + cr_d^t(P_{i,NEW,d}^t - x_{i,d}^t), \quad (3)$$

where c is the acceleration coefficients, r_d is a uniformly distributed random numbers in range $[0, 1]$. This search scheme make particle search more promising regions to find the global optimum. Research results showed that Eq. (3) was very useful for solving multimodal functions. In PSOCO, Eq. (3) still be adopted to update the particle velocity. Obviously, $P_{i,NEW}$ is a guiding vector. Hence, the focus of the PSOCO is to construct a good $P_{i,NEW}$.

3.1. Construction scheme of $P_{i,NEW}$

(1) arithmetical crossover: For each particle i , crossover operation is conducted on P_i and P_j to generate a new individual $V_i = [v_{i,1}, v_{i,2}, \dots, v_{i,D}]$ by the following equation:

$$V_{i,d} = r_{1,d}P_{i,d} + (1 - r_{1,d})P_{j,d}, \quad (4)$$

where $r_{1,d}$ is a random number uniformly distributed in $[0, 1]$. Firstly, the personal historical best particle swarm ($P_1, P_2, \dots, P_N$) do random permutations. As shown in the following matrix diagram:

$$\begin{bmatrix} P_{j1} & P_{j2} & \dots & P_{jN} \\ P_1 & P_2 & \dots & P_N \end{bmatrix}, \quad (5)$$

where $P_{j1}, P_{j2}, \dots, P_{jN}$ is a random permutation of $P_1, P_2, \dots, P_N$. Arithmetic crossover is conducted on the corresponding elements of the upper layer and the lower layer. Obviously, this method can produce offsprings that have good diversity. But, with the increment of iterations, it is important that the personal best P_i should do more exploitation in the neighbor space of the global best P_g . Hence, the crossover probability between P_i and P_g should be increased. Here, the replacement pressure can be controlled by R , which denotes that the number of P_j is replaced by P_g in the permutation swarm ($P_{j1}, P_{j2}, \dots, P_{jN}$). In order to balance the exploration and exploitation abilities, a dynamic adjustment strategy is designed for R . The R is obtained as

$$R = \lfloor t / \text{Maxit} \times N \rfloor, \quad (6)$$

where Maxit are the maximal iteration. Eq. (6) shows that R increases from 0 to the population size.

(2) differential evolution crossover: This crossover operation exchanges some components of the vector V_i with the target vector P_i to construct a trial vector U_i . Under this scheme, this operation can be expressed as:

$$U_{i,j} = \begin{cases} V_{i,j}, & \text{if } r_2 \leq CR \text{ or } j = j_{rand} \\ P_{i,j}, & \text{otherwise} \end{cases} \quad (7)$$

where CR is a user-defined parameter in the range $[0, 1]$, which controls the fraction of vector components inherited from V_i . r_2 is a random number uniformly distributed in $[0, 1]$. j_{rand} is a randomly chosen index

and it is randomly generated from the range $[1, D]$. Obviously, a small value CR can make the particle keep good diversity and make the algorithm do more exploration.

(3) competitive selection: the offspring U is created by applying two different crossover operations. A competition is made between U_i and P_i . As a result of competition selection, the exemplar with a better fitness will survive in this generation. This operation is described as:

$$P_{i,NEW} = \begin{cases} U_i, & \text{if } f(U_i) \leq f(P_i) \\ P_i, & \text{otherwise} \end{cases} \quad (8)$$

As shown in Eq. (8), this elitism scheme can ensure that the exemplar evolves toward a good direction at each iteration.

(4) replacement strategy of $P_{i,NEW}$: If the exemplar of a particle has stagnated in a certain number of generations, then this exemplar may be considered to be trapped in a local optimum. Here, the parameter G denotes the stagnated generation of the exemplar. Many studies have been done about the parameters G . According to these results, setting $G = 5$ or $G = 7$ is a good choice. In this paper, we set $G = 7$. The parameter setting can better balance the performance of PSOCO on lots of functions. In this case, replacement strategy is used to update the particle's exemplar. Firstly, $0.5 \times N$ exemplars are chosen at random. Then, these exemplars are sorted according to their fitness. Finally, the exemplar with the best fitness replaces the particle's current exemplar. By using this method, the particle changes the flying direction and the search position.

3.2. Procedures of the PSOCO

Compared with GPSO, the PSOCO adds a procedure of conducting $P_{i,NEW}$. The novelty of PSOCO lies in that it make full uses of crossover operations to process P_i of particles so as to breed the exemplars. Like GPSO, PSOCO is easy to be implemented and described. The procedure of PSOCO is presented as the following Algorithm 1.

Algorithm 1. PSOCO

```

1: Initialization the values of the necessary parameters;
2: Randomly initialize the positions and the velocities of  $N$  particles;
3: Evaluate the fitness values of the initial particle swarm;
4: Set  $X$  to be  $P_i$  and find  $P_g$ ;
5: while  $t < \text{Maxit}$  do
6: /* construction scheme of exemplars */
7: for  $i = 1$  to  $N$  do
8:   construct  $V_i$  by using Eq.(4);
9:   construct  $U_i$  by using Eq.(7);
10:  construct  $P_{i,NEW}$  by using Eq.(8);
11: end for
12: /*Velocity and Position adjusting*/
13: for  $i = 1$  to  $N$  do
14:   for  $d = 1$  to  $D$  do
15:     Update the velocity  $V_i$  of particle using Eq.(3);
16:   Update the position  $X_i$  of particle using Eq.(2);
17:   end for
18:   Evaluate  $f(X_i)$ ;
19:   Update  $P_i$  and  $P_g$ ;
20: end for
21: if  $P_{i,NEW}$  stops changing for  $G$  generations then
22:   Execute replacement strategy of  $P_{i,NEW}$ ;
23: end if
24:  $t = t + 1$ ;
25: end while

```

4. Experimental study

4.1. Benchmark functions and involved PSOs

In order to investigate the performance of PSOCO, three popular test suites consisting of 34 benchmark functions are used in the experiments.

Table 1
Fourteen test functions of the first test suite.

Test function	Search range	Optimal value
Sphere: $f_1(x) = \sum_{i=1}^D x_i^2$	$[-100, 100]^D$	0
SchwefelP2.22: $f_2(x) = \sum_{i=1}^D x_i + \prod_{i=1}^D x_i $	$[-10, 10]^D$	0
SumSquares: $f_3(x) = \sum_{i=1}^D (ix_i)^2$	$[-10, 10]^D$	0
Step: $f_4(x) = \sum_{i=1}^D [x_i + 0.5]^2$	$[-100, 100]^D$	0
Noise: $f_5(x) = \sum_{i=1}^D ix_i^4 + \text{random}[0, 1]$	$[-1.28, 1.28]^D$	0
Rosenbrock: $f_6(x) = \sum_{i=1}^{D-1} [100(x_{i+1} - x_i^2)^2 + (x_i - 1)^2]$	$[-10, 10]^D$	0
Schwefel: $f_7(x) = 418.9829 \times D - \sum_{i=1}^D x_i \sin(\sqrt{ x_i })$	$[-500, 500]^D$	0
Weierstrass: $f_8(x) = \sum_{i=1}^D (\sum_{k=0}^{kmax} [a_k \cos(2\pi b_k(x_i + 0.5))]) - D \sum_{k=0}^{kmax} [a_k \cos(2\pi b_k \times 0.5)]$, $a = 0.5$, $b = 3$, $kmax = 20$.	$[-0.5, 0.5]^D$	0
Rastrigin: $f_9(x) = \sum_{i=1}^D [x_i^2 - 10 \cos(2\pi x_i) + 10]$	$[-5.12, 5.12]^D$	0
Noncontinuous Rastrigin: $f_{10}(x) = \sum_{i=1}^D [y_i^2 - 10 \cos(2\pi y_i) + 10]$ if $ x_i \geq 0.5$ then $y_i = \text{round}(2x_i)/2$; else $y_i = x_i$	$[-5.12, 5.12]^D$	0
Ackley: $f_{11}(x) = -20 \exp(-0.2 \sqrt{1/D \sum_{i=1}^D x_i^2}) - \exp(1/D \sum_{i=1}^D \cos 2\pi x_i) + 20 + e$	$[-32, 32]^D$	0
Griewank: $f_{12}(x) = 1/4000 \sum_{i=1}^D x_i^2 - \prod_{i=1}^D \cos(x_i/\sqrt{i}) + 1$	$[-600, 600]^D$	0
Generalized Penalized1: $f_{13}(x) = \pi/D \{10 \sin^2(\pi y_i) + \sum_{i=1}^{D-1} (y_i - 1)^2 [1 + 10 \sin^2(\pi y_{i+1})] + (y_D - 1)^2\} + \sum_{i=1}^D u(x_i, 10, 100, 4)$	$[-50, 50]^D$	0
Generalized Penalized2: $f_{14}(x) = 1/10 \{ \sin^2(3\pi x_1) + \sum_{i=1}^{D-1} (x_i - 1)^2 [1 + \sin^2(3\pi x_{i+1})] + (x_D - 1)^2 [1 + \sin^2(2\pi x_D)] \} + \sum_{i=1}^D u(x_i, 5, 100, 4)$	$[-50, 50]^D$	0

Table 2
Ten test functions of the second test suite.

Test function	Search range	Optimal value
Rotated Ackley: $f_{15}(y) = -20 \exp(-0.2 \sqrt{1/D \sum_{i=1}^D y_i^2}) - \exp(1/D \sum_{i=1}^D \cos 2\pi y_i) + 20 + e$, where $y = M * x$, M is an orthogonal matrix	$[-32, 32]^D$	0
Rotated Griewank: $f_{16}(y) = 1/4000 \sum_{i=1}^D y_i^2 - \prod_{i=1}^D \cos(y_i/\sqrt{i}) + 1$ where $y = M * x$, M is an orthogonal matrix	$[-600, 600]^D$	0
Shifted Sphere: $f_{17}(z) = \sum_{i=1}^D z_i^2$ $z = x_i - o$, $o = [o_1, o_2, \dots, o_D]$: the shifted global optimum	$[-100, 100]^D$	0
shifted Weierstrass: $f_{18}(z) = \sum_{i=1}^D (\sum_{k=0}^{kmax} [a_k \cos(2\pi b_k(z_i + 0.5))]) - D \sum_{k=0}^{kmax} [a_k \cos(2\pi b_k \times 0.5)]$, $a = 0.5$, $b = 3$, $kmax = 20$. $z = x - o$, $o = [o_1, o_2, \dots, o_D]$: the shifted global optimum	$[-0.5, 0.5]^D$	0
Shifted Rastrigin: $f_{19}(z) = \sum_{i=1}^D [z_i^2 - 10 \cos(2\pi z_i) + 10]$, $z = x - o$ $o = [o_1, o_2, \dots, o_D]$: the shifted global optimum	$[-5.12, 5.12]^D$	0
Shifted Noncontinuous Rastrigin: $f_{20}(x) = \sum_{i=1}^D [y_i^2 - 10 \cos(2\pi y_i) + 10]$ if $ z_i \geq 0.5$ then $y_i = \text{round}(2z_i)/2$; else $y_i = z_i$ where $z = x - o$; $o = [o_1, o_2, \dots, o_D]$: the shifted global optimum	$[-5.12, 5.12]^D$	0
Shifted Ackley: $f_{21}(z) = -20 \exp(-0.2 \sqrt{1/D \sum_{i=1}^D z_i^2}) - \exp(1/D \sum_{i=1}^D \cos 2\pi z_i) + 20 + e$ $z = x_i - o$, $o = [o_1, o_2, \dots, o_D]$: the shifted global optimum	$[-32, 32]^D$	0
Shifted Griewank: $f_{22}(z) = 1/4000 \sum_{i=1}^D z_i^2 - \prod_{i=1}^D \cos(z_i/\sqrt{i}) + 1$ $z = x_i - o$, $o = [o_1, o_2, \dots, o_D]$: the shifted global optimum	$[-600, 600]^D$	0
Shifted Generalized Penalized1: f_{23}	$[-50, 50]^D$	0
Shifted Generalized Penalized2: f_{24}	$[-50, 50]^D$	0

The first set of benchmark functions includes 14 classical benchmark functions. Table 1 describes 14 classical benchmarks. Detailed descriptions of these benchmarks can be found in Yao et al. (1999). In this test suite, $f_1 - f_6$ are unimodal functions. These functions are often used to test the convergence feature of the algorithms. Next, $f_7 - f_{14}$ are multimodal functions. These eight functions are used to test the exploration ability (avoid local optima) of the algorithms. The second set of benchmark functions consists of 10 functions. Table 2 gives a detailed description about these functions. The third test set consists of ten functions from the CEC 2013 test set. The list of functions used is given in Table 3 and more details can be found in Liang et al. (0000). For the first test set and the second test set, eight PSOs are used to verify the efficiency of the proposed PSOCO. The parameter configurations of variant PSO algorithms are listed in detail in Table 4.

In this experiment, the swarm size is set to be 50. The dimension of test functions is 30. The maximum number of function evaluations

(FEs) is set to be 300 000 for each test function. On each benchmark function, 25 independent trials are run. The mean results and the standard deviation are recorded for comparison.

4.2. Some explanations about PSOCO

This section mainly focuses on the influences of parameters to find their proper values. We adopt the following method for the investigation of the parameters. When the parameters investigated are changed and the others remain unchanged, the performance of PSOCO is observed. All the benchmark functions of the first test suite in Table 1 are selected.

4.2.1. Effects of different combinations of w and c

In many algorithms, two values are often used by parameter w . One is reduced linearly from 0.9 to 0.4. Another is 0.7298 (Mendes et al., 2004; Wang et al., 2008; Hao et al., 2007; Gong et al., 2016). The

Table 3

Ten test functions of the third test suite.

Function	Type	Name	NO. in CEC 2013
f_{25}	Unimodal function	Different powers function	NO.6
f_{26}	Basic multimodal function	Rastrigins function	NO.11
f_{27}		Schwefel's function	NO.14
f_{28}		Lunacek Bi_Rastrigin function	NO.17
f_{29}		Expanded Griewanks plus Rosenbrocks function	NO.19
f_{30}	Composition function	Composition function 1 (Rotated)	NO.21
f_{31}		Composition function 2 (Unrotated)	NO.22
f_{32}		Composition function 6 (Rotated)	NO.26
f_{33}		Composition function 7 (Rotated)	NO.27
f_{34}		Composition function 8 (Rotated)	NO.28

Table 4

Parameters settings.

Algorithm	Parameters settings	Reference
GPSO	$w : 0.9-0.4, c_1 = c_2 = 2.0$	Shi and Eberhart (1998)
FIPS	$\chi = 0.7298, \Sigma c_i = 4.1$	Mendes et al. (2004)
PSOPC	$w : 0.792, c_1 = c_2 = 1.887, crossiter = 2$	Chen (2012)
CLPSO	$w : 0.9-0.2, c = 1.49445, m = 7$	Liang et al. (2006)
FDR-PSO	$w : 0.9-0.4, c_1 = c_2 = 1.0, c_3 = 2$	Peram et al. (2003)
MNPSO	$w = 0.7298, c_1 = c_2 = 1.49618$	Wang et al. (2008)
MPSO	$\chi = 0.7298, c_1 = c_2 = 2.05$	Hao et al. (2007)
SaDPSO	$w \in [0.4, 0.9], c_1 = c_2 = 1.49445$	Liang et al. (2015)
PSOCO	$w : 0.7298, c = 1.49618, G = 7, CR = 0.05$	–

most common values of the acceleration coefficient c are 2 and 1.49618 (Wang et al., 2008; Gong et al., 2016). Here, we mainly check the influences of different combinations of these two parameters. It can be shown from Table 5 that the combination of $w = 0.7298$ and $c = 1.49618$ performs best among all four combinations for most of the functions.

4.2.2. Influences of parameter CR

Different CR has been chosen for the simulation. For each selected CR, 25 runs are performed and the means results are given in Table 6. As can be seen from Table 6, when CR is large, PSOCO has better performance for unimodal functions f_1 and f_3 . When CR is small, PSOCO has better performance for multimodal problems. In conclusion, with a CR = 0.05, PSOCO have a better performance on lots of unimodal and multimodal problems. According to the above results, we finally set CR = 0.05 for PSOCO.

4.3. Discussion of experimental results on first test suite

The mean values, best values and the standard deviations (SD) obtained by several different algorithms are shown in Table 7, where the best results are marked in bold. The experimental results of 25 trials on each function show that PSOCO has a excellent ability to find the

global optimum on most of the tested functions. PSOCO obtains the best solutions on the functions $f_4, f_8, f_9, f_{10}, f_{12}, f_{13}, f_{14}$.

(1) Unimodal functions

Because FDR-PSO adds a nearest-better particle to guide the flying of the particle, it has the best performance on unimodal functions f_1, f_2, f_3 . Furthermore, FDR-PSO obtains high-quality solutions on f_1, f_2, f_3 . The main reason is that the globally best particle is used to guide all the particles. Although, the result of PSOCO is not better than that of FDR-PSO on f_1, f_2, f_3, f_5 and f_6 , PSOCO converges to the global optimum 0 on f_4 . SaDPSO has the best performance on f_5 and f_6 . Only SaDPSO can achieve the global optimum and obtain high-quality mean solutions on f_6 . As shown in Table 7, SaDPSO has the best overall performance for all basic unimodal functions. The evolutionary curves of GPSO and PSOCO on f_1, f_2 are given in Fig. 1. The figure shows the convergence characteristics of f_1 and f_2 . As can be seen from Fig. 1 that PSOCO have fast convergence rate at the early stage of optimization. At the later search process, PSOCO have a strong ability of refined search compared with GPSO. However, compared with GPSO, PSOPC, FIPS, CLPSO and MPSO, PSOCO can still achieve better solution on f_1, f_2, f_3 . In short, PSOCO has a balance performance compared with the other eight PSO variants for unimodal functions.

(2) Multimodal functions

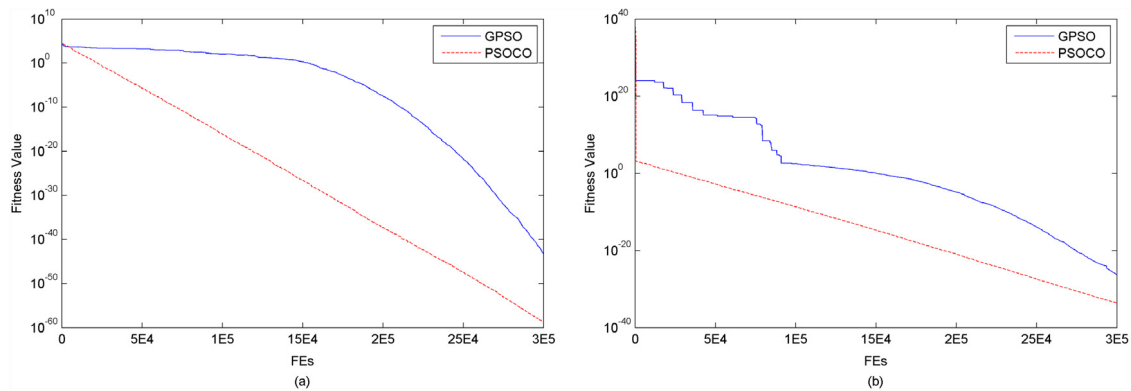
Owing to the complex of real-world problems, the shape of the fitness landscape is not known. It is very important that an algorithm is good at solving multimodal functions, which contain a number of local optima. In solving multimodal functions, some PSO algorithms has disadvantages because these algorithms are easy to make premature convergence. As can be seen from Table 7, the performances of GPSO, PSOPC, FDR-PSO, FIPS, MNPSO, MPSO, SaDPSO are worse than that of CLPSO and PSOCO. The global optimum of function f_7 is near the boundary of the search space, so it is difficult for some PSO algorithms to find the global optimum. As shown in Table 7, most algorithms obtain relatively bad results on f_7 . However, CLPSO and PSOCO are still the best algorithms on f_7 . The functions f_9 and f_{10} are very complex multimodal problems. In order to better solve these problems, the

Table 5Experimental performances with different parameter combinations of w and c .

Fun	$w : 0.9-0.4$	$w : 0.9-0.4$	$w : 0.7298$	$w : 0.7298$
	$c = 2$	$c = 1.49618$	$c = 2$	$c = 1.49618$
f_1	4.56e-01 ± 6.74e-01	9.10e-08 ± 2.54e-07	2.96e-51 ± 4.37e-51	7.67e-59 ± 7.43e-59
f_2	6.12e-02 ± 4.30e-02	2.15e-05 ± 2.74e-05	7.91e-30 ± 4.75e-31	1.51e-35 ± 9.30e-36
f_3	1.07e00 ± 1.44e00	6.99e-08 ± 9.50e-08	1.41e-51 ± 7.03e-51	5.51e-58 ± 8.82e-58
f_4	9.60e-01 ± 3.98e00	0 ± 0	0 ± 0	0 ± 0
f_5	2.18e-02 ± 1.97e-02	1.50e-02 ± 4.70e-03	9.30e-03 ± 1.80e-03	1.90e-02 ± 5.10e-03
f_6	8.73e+01 ± 2.49e+01	2.85e+01 ± 3.24e+01	1.59e+01 ± 2.66e+01	1.01e+01 ± 1.80e+01
f_7	4.73e+01 ± 4.21e+01	5.28e-04 ± 1.58e-04	3.82e-04 ± 9.27e-12	3.82e-04 ± 4.36e-12
f_8	1.11e-01 ± 6.56e-02	1.91e-02 ± 2.96e-02	0 ± 0	0 ± 0
f_9	9.14e-03 ± 1.59e-02	4.34e-07 ± 8.31e-07	0 ± 0	0 ± 0
f_{10}	2.51e-03 ± 4.50e-03	1.22e-06 ± 1.54e-06	0 ± 0	0 ± 0
f_{11}	8.42e-02 ± 1.38e-01	1.40e-05 ± 8.99e-05	2.62e-14 ± 4.74e-15	2.41e-14 ± 4.60e-15
f_{12}	3.20e-01 ± 2.68e-01	7.80e-03 ± 8.40e-03	0 ± 0	0 ± 0
f_{13}	4.13e-01 ± 6.67e-01	4.46e-07 ± 5.50e-07	1.57e-32 ± 8.38e-48	1.57e-32 ± 8.38e-48
f_{14}	7.46e-01 ± 5.11e-01	5.37e-07 ± 7.50e-07	1.35e-32 ± 5.59e-48	1.35e-32 ± 5.59e-48

Table 6Performance of PSOCO with different CR .

Fun	Term	CR					
		0.025	0.05	0.1	0.2	0.4	0.8
f_1	Mean	6.73e-49	7.67e-59	3.53e-69	8.52e-77	1.50e-86	1.15e-91
	SD	3.10e-49	7.43e-59	4.21e-69	1.34e-76	7.34e-86	8.22e-91
f_2	Mean	2.40e-29	1.51e-35	3.54e-42	3.66e-47	6.19e-29	1.06e-12
	SD	9.97e-30	9.30e-36	6.85e-43	5.78e-47	1.90e-28	3.08e-11
f_3	Mean	3.68e-48	5.51e-58	1.70e-69	2.56e-77	1.14e-85	5.24e-93
	SD	1.47e-48	8.82e-58	1.13e-68	2.16e-76	3.07e-85	2.41e-92
f_4	Mean	0	0	0	8.00e-02	6.80e-01	4.32e00
	SD	0	0	0	2.77e-01	6.19e-01	3.42e00
f_5	Mean	2.53e-02	1.90e-02	3.12e-02	1.12e-01	2.91e-01	5.22e-01
	SD	7.60e-03	5.10e-03	1.01e-02	3.06e-02	6.96e-02	1.35e-01
f_6	Mean	2.03e00	1.01e+01	4.35e+01	3.52e+01	3.77e+01	2.56e+01
	SD	6.41e00	1.80e+01	2.95e+01	2.25e+01	2.32e+01	1.57e+01
f_7	Mean	3.82e-04	3.82e-04	9.47e00	1.42e+01	1.24e+02	4.03e+03
	SD	9.21e-12	4.36e-12	3.27e+01	3.93e+01	1.08e+01	5.74e+02
f_8	Mean	0	0	0	0	4.61e-02	5.92e-01
	SD	0	0	0	0	4.84e-02	7.13e-01
f_9	Mean	0	0	0	1.19e-01	1.19e+01	2.92e+01
	SD	0	0	0	3.30e-01	1.81e00	7.06e00
f_{10}	Mean	0	0	8.00e-02	5.20e-01	7.16e00	2.16e+01
	SD	0	0	2.77e-01	5.86e-01	6.12e00	5.19e00
f_{11}	Mean	3.93e-14	2.41e-14	2.15e-14	3.73e-02	1.43e00	2.29e00
	SD	2.40e-15	4.60e-15	1.54e-15	2.58e-01	3.92e-01	7.06e00
f_{12}	Mean	0	0	0	2.96e-04	1.60e-03	4.30e-03
	SD	0	0	0	1.50e-03	3.30e-03	8.06e-03
f_{13}	Mean	1.57e-32	1.57e-32	1.57e-32	3.64e-32	7.04e-01	4.08e00
	SD	8.38e-48	8.38e-48	8.38e-48	3.94e-32	1.28e+01	2.56e+01
f_{14}	Mean	1.35e-32	1.35e-32	1.35e-32	1.35e-32	1.04e-01	5.76e-01
	SD	5.59e-48	5.59e-48	5.59e-48	5.59e-48	1.53e-01	1.73e00

**Fig. 1.** Convergence progresses of GPSO and PSOCO on unimodal functions. (a) f_1 , (b) f_2 .

mechanism for keeping enough diversity is critical to many algorithms. Similarly, the five algorithms, namely, GPSO, FDR-PSO, FIPS, SaDPSO and PSOPC, are trapped into local optima. Owing to maintain the diversity of the swarm, PSOCO is very effective on multimodal function. PSOCO outperforms all the other PSOs on functions $f_8, f_9, f_{10}, f_{12}, f_{13}$ and f_{14} . Especially, only PSOCO can reach the optimum value 0 on the Griewank function (f_{12}). Obviously, GPSO and FDR-PSO are easy to be attracted by particle with the best fitness. Thus, GPSO and FDR-PSO are difficult to escape the current best space which is more likely to a local optimum. Fig. 2 also shows the advantages of PSOCO for solving multimodal functions. PSOCO lets the personal historically best particle (P_i) cross with each other to construct a new vector U_i , and then the probability CR is used to control information sharing of U_i and P_i . Finally, a promising guidance vector P_{iNEW} is constructed. These mechanism can avoid local optima for the whole swarm.

(3) Result comparisons of search speed and reliability

Table 8 shows the convergence speed of each PSO by giving the fitness evaluations (FEs) and successful ratio (SR) that stands for the percentage the successful runs in the 25 independent runs. Seen from Table 8, MNPSO converges the fastest on $f_1, f_2, f_3, f_4, f_6, f_8, f_9$ and f_{11} in terms of FEs. However, on the aspect of reliability, MNPSO has the worst performance on f_7, f_{13} and f_{14} . On the aspect of convergence speed, PSOCO converges the fastest on $f_7, f_{10}, f_{12}, f_{13}$ and f_{14} . We can observe from Table 8 that PSOCO is also efficient in solving the first suite problem. Compared with other PSOs, PSOCO reaches the acceptive solutions on all functions except f_5 with fewer FEs. Furthermore, PSOCO reaches the ratio of 100% on 13 out of 14 tested functions. The experimental results show that PSOCO is very competitive on the reliability and convergence speed. In short, PSOCO is an effective combination of crossover operators and PSO algorithms.

Table 7
Solution quality comparison among nine algorithms.

Fun	Term	GPSO	FIPS	PSOPC	CLPSO	FDR-PSO	MNPSO	MPSO	SaDPSO	PSOCO
f_1	Mean	4.98e-40	3.37e-15	7.06e-16	4.84e-24	1.60e-139	2.92e-63	6.72e-45	1.78e-65	7.67e-59
	SD	1.63e-39	1.79e-15	1.63e-15	2.55e-24	6.29e-139	1.22e-62	1.94e-44	5.35e-65	7.43e-59
	Best	1.70e-43	6.37e-16	1.04e-20	1.58e-24	1.35e-149	5.45e-69	5.28e-49	2.98e-75	8.80e-60
f_2	Mean	3.80e-27	1.33e-09	3.85e-08	3.02e-15	3.37e-59	1.27e-39	3.01e-25	5.72e-35	1.51e-35
	SD	1.17e-26	3.37e-10	6.02e-08	1.05e-15	1.68e-58	6.05e-39	4.19e-25	2.50e-34	9.30e-36
	Best	1.90e-29	8.63e-10	5.82e-10	1.48e-15	8.71e-73	2.47e-43	1.29e-26	2.11e-39	4.86e-36
f_3	Mean	1.16e-40	4.69e-15	1.13e-14	2.46e-24	1.81e-135	2.82e-62	2.75e-25	9.59e-64	5.51e-58
	SD	2.74e-40	2.22e-15	3.88e-14	1.26e-24	9.07e-135	9.93e-62	1.38e-24	3.11e-63	8.82e-58
	Best	3.38e-43	8.17e-15	2.49e-19	8.46e-24	3.12e-149	1.06e-69	2.81e-46	6.00e-75	5.43e-59
f_4	Mean	8.00e-2	0	3.90e+01	0	8.00e-02	0	0	0	0
	SD	2.77e-1	0	4.57e+01	0	2.77e-01	0	0	0	0
	Best	0	0	7.00e00	0	0	0	0	0	0
f_5	Mean	1.20e-02	5.10e-3	6.52e-02	5.60e-03	3.70e-03	1.19e-02	3.60e-03	3.40e-03	1.90e-02
	SD	4.10e-03	1.40e-3	2.51e-02	1.50e-03	1.70e-03	5.20e-03	1.30e-03	1.30e-03	5.10e-03
	Best	6.00e-03	1.80e-03	3.01e-02	2.5e-03	1.20e-03	2.50e-03	1.40e-03	1.70e-03	7.10e-03
f_6	Mean	2.83e+01	2.43e+01	2.02e+01	1.02e+01	3.03e00	1.54e+01	2.98e+01	7.12e-11	1.01e+01
	SD	2.75e+01	3.29e-01	2.01e+01	5.62e00	2.58e00	1.46e+00	2.64e+01	2.52e-11	1.80e+01
	Best	1.19e00	2.37e+01	2.50e00	2.67e00	7.26e-07	1.29e+01	2.70e-03	4.83e-11	2.02e-02
f_7	Mean	1.02e+03	1.36e00	8.03e+03	3.82e-04	3.18e+03	3.14e+03	3.51e+02	1.72e+03	3.82e-04
	SD	2.56e+02	3.61e00	2.24e+02	0	4.25e+03	4.69e+02	1.86e+02	3.52e+02	4.36e-12
	Best	5.92e+02	3.82e-04	7.68e+03	3.82e-04	2.17e+03	2.20e+03	1.18e+02	9.48e+02	3.82e-04
f_8	Mean	6.00e-02	7.68e-02	6.55e00	0	1.81e-01	0	0	0	0
	SD	3.00e-01	7.87e-02	1.78e00	0	4.99e-01	0	0	0	0
	Best	0	5.22e-04	3.62e00	0	0	0	0	0	0
f_9	Mean	2.47e+01	6.17e+01	2.02e+01	7.15e-14	2.39e+01	0	1.99e-01	2.27e+00	0
	SD	8.66e00	1.27e+01	4.30e+01	8.22e-14	6.33e00	0	4.06e-01	1.51e+00	0
	Best	1.19e+01	3.72e+01	3.98e00	1.78e-15	9.95e00	0	0	0	0
f_{10}	Mean	1.34e+01	5.93e+01	6.95e+01	1.14e-12	4.12e+01	0	8.00e-02	5.44e+00	0
	SD	1.06e+01	8.73e00	5.81e+01	8.70e-13	5.89e00	0	2.77e-01	2.10e+00	0
	Best	3.13e-06	3.95e+01	6.00e00	2.18e-13	3.00e00	0	0	2.00e+00	0
f_{11}	Mean	3.04e-14	1.27e-08	4.24e00	5.12e-13	2.10e-14	2.13e-15	7.82e-15	7.11e-15	2.41e-14
	SD	5.13e-15	3.72e-09	9.17e-01	1.26e-13	6.15e-15	1.78e-15	2.51e-15	0	4.60e-15
	Best	1.42e-14	7.49e-09	2.32e00	3.06e-13	1.42e-14	0	3.55e-15	7.11e-15	1.51e-14
f_{12}	Mean	1.22e-02	2.41e-06	2.17e-02	2.73e-15	1.35e-02	7.10e-04	8.20e-03	2.96e-04	0
	SD	1.42e-02	8.70e-06	2.12e-02	6.87e-15	1.24e-02	1.70e-03	1.37e-02	1.52e-03	0
	Best	0	3.76e-14	2.22e-16	0	0	0	0	0	0
f_{13}	Mean	1.77e-32	6.32e-15	6.93e00	2.66e-24	1.57e-32	1.15e+03	5.19e+01	1.57e-32	1.57e-32
	SD	9.86e-33	4.11e-15	1.72e+01	1.72e-24	8.38e-48	1.34e+03	1.82e+02	8.38e-48	8.38e-48
	Best	1.57e-32	2.06e-15	5.61e-19	2.92e-25	1.57e-32	3.20e+01	1.57e-32	1.57e-32	1.57e-32
f_{14}	Mean	8.79e-04	7.17e-13	8.42e-01	1.70e-21	4.39e-04	2.32e+03	7.52e+01	1.35e-32	1.35e-32
	SD	3.00e-03	1.54e-12	2.63e00	3.56e-21	2.20e-03	1.16e+03	2.60e+02	5.59e-48	5.59e-48
	Best	1.35e-32	1.25e-15	4.60e-17	2.33e-24	1.35e-32	1.35e+01	1.35e-32	1.35e-32	1.35e-32

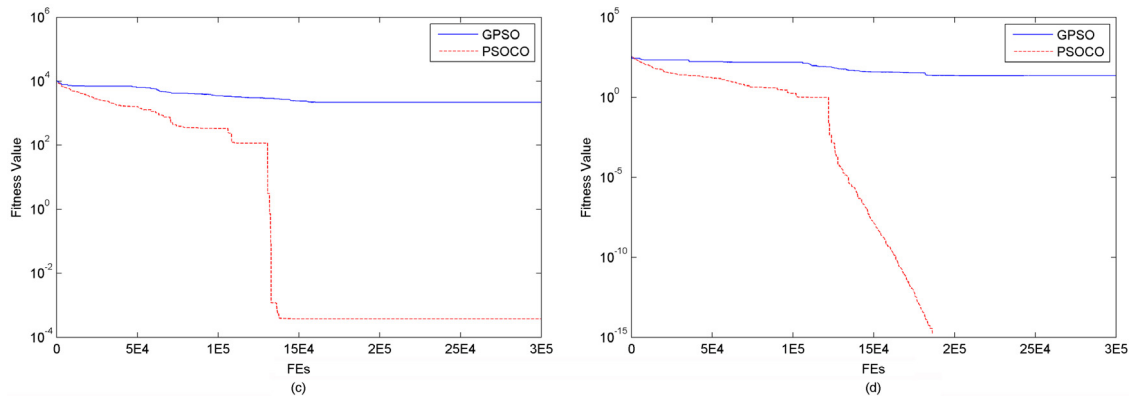


Fig. 2. Convergence progresses of GPSO and PSOCO on multimodal functions. (c) f_7 , (d) f_9 .

4.4. Further performance on second test suite

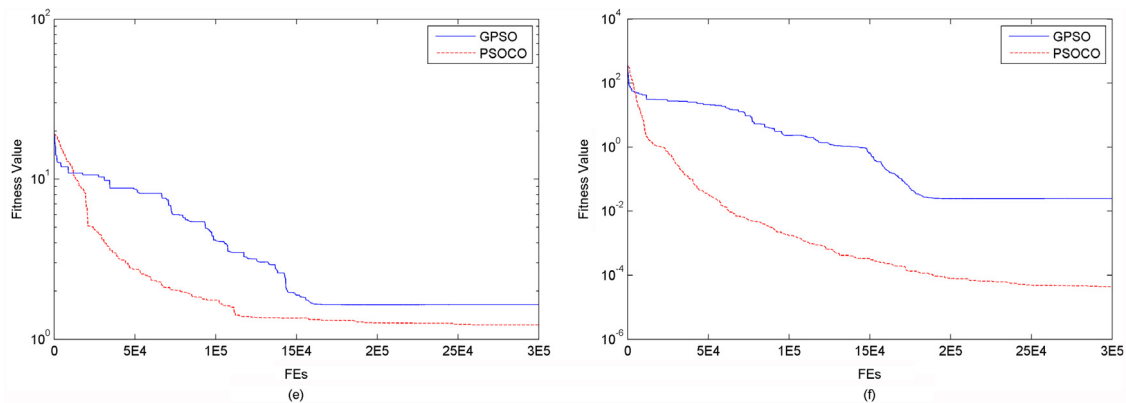
This section further tests PSOCO under different complex functions. The second test suite includes two rotated functions and eight shifted functions which are very difficult to be optimized. These functions

can further check the performance of PSOCO. To rotated functions, an orthogonal matrix M will be generated. In order to test the performance of the algorithm fairly, a new rotation matrix M is computed before each run of the trials. The mean and standard deviations of the results obtained by nine algorithms are shown in Table 9. Obviously, owing to

Table 8

Convergence speed and algorithm reliability comparisons among nine algorithms.

Fun	Term	GPSO	FIPS	PSOPC	CLPSO	FDR-PSO	MNPSO	MPSO	SaDPSO	PSOCO	ACCEPT
f_1	FES	196 792	161 968	136 142	141 096	119 663	42 517	62 156	57 454	51 900	1e–6
	SR%	100	100	100	100	100	100	100	100	100	
f_2	FES	202 002	217 368	215 720	169 388	83 294	56 266	87 854	76 355	68 180	1e–6
	SR%	100	100	92	100	100	100	100	100	100	
f_3	FES	197 058	164 012	138 404	119 590	139 999	43 876	104 608	58 154	53 350	1e–6
	SR%	100	100	100	100	100	100	100	100	100	
f_4	FES	150 806	64 598	–	79 645	88 073	18 505	30 452	35 025	23 200	0
	SR%	92	100	0	100	92	100	100	100	100	
f_5	FES	281 685	171 310	–	200 285	148 206	185 834	128 024	196 076	270 016	0.01
	SR%	28	100	0	100	100	48	100	100	24	
f_6	FES	166 510	51 086	26 700	1 115 074	83 294	12 674	53 352	27 652	33 892	100
	SR%	100	100	100	100	100	100	100	100	100	
f_7	FES	117 254	88 636	–	37 719	–	–	40 354	88 480	37 332	2e3
	SR%	100	100	0	100	0	0	100	72	100	
f_8	FES	176 086	168 380	–	107 718	104 380	27 634	70 902	64 755	38 944	0.1
	SR%	100	100	0	100	100	100	100	100	100	
f_9	FES	139 666	102 866	148 198	45 688	52 618	9357	11 252	20 254	10 816	100
	SR%	100	100	96	100	100	100	100	100	100	
f_{10}	FES	160 078	75 114	67 436	29 770	46 080	10 930	14 044	16 750	7320	100
	SR%	100	100	60	100	100	100	100	100	100	
f_{11}	FES	219 210	238 496	–	186 758	136 002	69 360	92 220	78 458	79 784.1e–6	
	SR%	100	100	0	100	100	100	100	100	100	
f_{12}	FES	199 079	196 714	129 083	174 764	121 572	107 494	76 470	74 940	60 176	1e–6
	SD	28	96	12	100	36	76	72	96	100	
f_{13}	FES	205 076	164 912	177 106	140 142	113 520	–	149 153	62 958	51 976	1e–6
	SR%	100	100	60	100	100	0	88	100	100	
f_{14}	FES	207 435	161 638	213 993	141 771	116 717	–	119 956	60 254	54 808	1e–6
	SR%	80	100	12	100	92	0	84	100	100	

**Fig. 3.** Convergence progresses of GPSO and PSOCO on rotated functions. (e) f_{15} , (f) f_{16} .

the coordinate rotation of particle, all the PSO algorithms are affected. MNPSO surpasses all other algorithms on f_{15} and f_{16} and achieves high quality solutions. Moreover, Fig. 3 shows that PSOCO can perform better than GPSO on these two rotated functions.

On shifted functions, PSOCO performs best on functions f_{17} , f_{18} , f_{19} , f_{20} and f_{22} , 5 out of the 8 functions. Table 9 shows that only PSOCO, SaDPSO and FDR-PSO can achieve the global optimum 0 on f_{17} and only PSOCO and CLPSO can achieve the global optimum 0 on f_{18} and f_{19} . The evolutionary curves of GPSO and PSOCO on f_{18} , f_{21} are given in Fig. 4. The curves of PSOCO descend much faster than that of GPSO, suggesting better performance of PSOCO in search speed. Especially, only PSOCO can achieve the optimum value 0 on f_{22} . Note that it can be observed that CLPSO is the second best on shifted functions. In conclusion, PSOCO is competitive PSO variant compared with other PSO variants for some benchmark functions.

4.5. Non-parameter statistic significant

To further analyze whether there is significant difference between the results of each algorithm on benchmark functions (f_1 – f_{24}), statistical t -tests were performed. The number of benchmark functions showing that PSOCO is significantly better than, almost the same as, and significantly worse than the other algorithms is shown in Table 10. The level of significance is 0.05. The “Merit” score was calculated by subtracting the “Worse” score from the “Better” score. As can be seen from Table 10, PSOCO is better than the compared algorithms. This demonstrates that the PSOCO has remarkable competitive advantages.

4.6. Comparison between PSOCO with other evolutionary algorithms

To further verify the performance of PSOCO, several state-of-the-art metaheuristic algorithms with different inspiring mechanisms are used

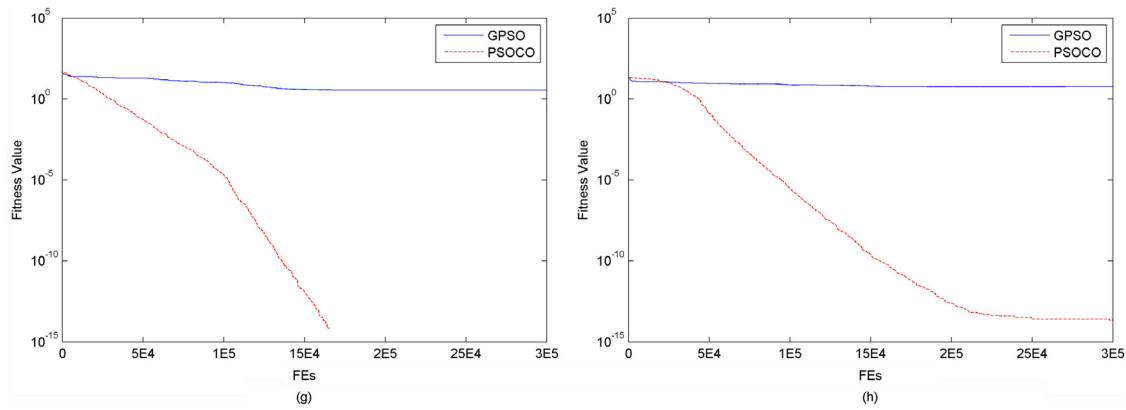


Fig. 4. Convergence progresss of GPSO and PSOCO on shifted functions. (g) f_{18} , (h) f_{21} .

Table 9

Results of nine algorithms on the second test suite.

Fun	Term	GPSO	FIPS	PSOPC	CLPSO	FDR-PSO	MNPSO	MPSO	SaDPSO	PSOCO
f_{15}	Mean	1.04e00	1.67e-08	7.26e00	4.01e-07	4.20e-01	2.84e-15	4.87e-01	2.04e-06	1.73e-01
	SD	7.19e-01	3.94e-09	2.71e00	1.07e-06	6.70e-01	1.45e-15	7.57e-01	5.75e-06	4.72e-01
	Best	2.13e-14	1.03e-08	2.89e00	1.87e-11	1.42e-14	0	7.11e-15	3.55e-15	3.54e-14
f_{16}	Mean	1.39e-02	1.70e-07	5.62e-02	1.04e-06	1.22e-02	0	6.80e-03	1.70e-03	6.94e-04
	SD	1.25e-02	6.97e-07	4.91e-02	1.88e-06	1.60e-02	0	1.02e-02	4.90e-03	1.20e-03
	Best	0	3.66e-12	1.67e-15	5.27e-11	0	0	0	0	7.31e-15
f_{17}	Mean	1.69e+01	4.37e-15	2.65e-14	1.69e-25	0	4.49e-29	1.01e+03	0	0
	SD	2.78e+01	2.19e-15	1.11e-13	1.42e-25	0	7.87e-29	1.43e+03	0	0
	Best	0	1.70e-15	6.06e-19	2.69e-25	0	0	0	0	0
f_{18}	Mean	5.49e00	1.27e00	1.73e+01	0	3.19e00	2.04e+01	7.62e00	1.82e-01	0
	SD	3.21e00	7.11e-01	3.88e00	0	1.96e00	7.80e+01	3.52e00	3.39e-01	0
	Best	0	1.93e-01	9.74e00	0	2.64e-04	1.18e+01	1.44e00	0	0
f_{19}	Mean	2.05e+01	6.08e+01	1.57e+02	0	3.03e+01	9.13e+01	5.22e00	5.41e00	0
	SD	5.50e00	1.20e+01	6.72e+01	0	7.56e00	2.80e+01	7.15e00	2.60e00	0
	Best	1.09e+01	3.31e+01	5.77e+01	0	1.69e+01	4.97e+01	0	9.95e-01	0
f_{20}	Mean	1.12e+01	5.94e+01	1.82e+02	4.00e-02	1.30e+01	5.66e+01	1.92e00	8.25e00	4.00e-02
	SD	5.33e00	1.31e+01	5.01e+01	2.00e-01	7.42e00	3.09e+01	2.27e00	2.71e00	2.00e-01
	Best	1.00e00	3.95e+01	9.84e+01	0	3.00e00	6.00e00	0	3.24e00	1.78e-15
f_{21}	Mean	1.33e00	1.44e-06	6.53e00	6.81e-13	1.49e-14	3.49e00	7.35e00	8.95e-15	2.42e-14
	SD	3.99e00	7.08e-06	2.52e00	5.21e-13	6.15e-15	6.33e00	3.24e00	3.10e-15	3.71e-15
	Best	2.13e-14	1.05e-08	2.74e00	1.70e-13	7.11e-15	7.11e-15	2.84e-14	7.11e-15	1.87e-14
f_{22}	Mean	5.68e-01	1.79e-08	2.95e-01	6.22e-16	1.01e-02	2.35e-02	1.64e+01	6.24e-13	0
	SD	5.88e-01	5.23e-08	7.49e-01	6.57e-15	1.05e-02	5.49e-02	1.41e+01	3.12e-12	0
	Best	0	2.64e-14	3.33e-16	0	0	0	1.14e00	0	0
f_{23}	Mean	3.58e-01	3.83e-14	8.95e+02	6.30e-03	5.90e-03	8.55e+02	2.53e+03	2.80e-03	1.36e-03
	SD	1.00e00	4.54e-14	8.55e+02	8.22e-14	1.54e-02	1.34e+03	1.02e+04	9.20e-03	5.07e-03
	Best	1.57e-32	8.35e-15	2.67e-12	1.24e-26	1.57e-32	1.57e-32	1.57e-32	1.57e-32	1.57e-32
f_{24}	Mean	6.04e-01	2.91e-12	8.04e+01	8.30e-03	2.44e-02	8.15e+02	4.97e+04	1.26e-02	1.26e-02
	SD	1.10e00	7.80e-12	1.21e+02	8.70e-13	4.59e-02	1.90e+03	1.62e+05	2.61e-02	2.70e-02
	Best	1.35e-32	5.13e-15	1.10e-02	1.44e-25	1.35e-32	1.35e-32	1.35e-32	1.35e-32	1.35e-32

Table 10

The t -test results between PSOCO and other algorithms.

Term	GPSO	FIPS	PSOPC	CLPSO	FDR-PSO	MNPSO	MPSO	SaDPSO
Better	15	14	19	10	9	12	13	9
Same	9	6	5	11	10	6	9	9
Worse	0	4	0	3	5	6	2	6
Merit	15	10	19	7	4	6	11	3

for comparison. These algorithms are DE (Storn and Price, 1997), SPSO (Zambrano-Bigiarini et al., 2013), CMA-ES (Hansen and Ostermeier, 2001), EFWA (Zheng et al., 2013), AFWA (Li et al., 2014), dynFWA (Zheng et al., 2014) and GFWA (Li et al., 2017). All the algorithms are run 51 times for each function. Some results can be found in Li et al. (2017). The mean of eight algorithms on the third test suite are compared and listed in Table 11. PSOCO beats other algorithms on f_{26} , f_{27} , f_{28} , f_{29} , f_{31} , f_{32} and f_{34} . CMA-ES performs well on f_{25} and f_{33} .

According to experimental results, the PSOCO performs the best among these eight algorithms on the third test suite.

5. Conclusion

In this paper, a new optimization scheme for PSO namely PSOCO has been proposed. PSOCO adopts two different crossover operations. One is arithmetical crossover, the other is differential evolution crossover. The

Table 11

Results of eight algorithms on the third test suite.

Fun	Term	DE	SPSO	CMA-ES	EFWA	AFWA	dynFWA	GFWA	PSOCO
f_{25}	Mean	2.48e−03	5.42e−04	0.00e+00	7.90e−02	6.04e−04	5.42e−04	1.55e−03	2.94e−13
f_{26}	Mean	7.88e+01	1.05e+02	4.00e+02	4.22e+02	1.05e+02	1.07e+02	7.50e+01	6.80e−14
f_{27}	Mean	2.38e+03	3.99e+03	4.94e+03	4.16e+03	2.97e+03	2.95e+03	3.49e+03	1.96e+01
f_{28}	Mean	9.29e+01	1.16e+02	7.44e+02	3.10e+02	1.45e+02	1.48e+02	8.49e+01	3.10e+01
f_{29}	Mean	4.51e+00	9.51e+00	3.54e+00	1.23e+01	6.92e+00	6.87e+00	5.08e+00	7.57e−01
f_{30}	Mean	3.20e+02	3.09e+02	3.44e+02	3.24e+02	3.16e+02	2.92e+02	2.59e+02	2.63e+02
f_{31}	Mean	1.72e+03	4.30e+03	7.97e+03	5.75e+03	3.45e+03	3.41e+03	4.27e+03	1.19e+02
f_{32}	Mean	2.52e+02	2.86e+02	3.29e+02	3.21e+02	2.73e+02	2.62e+02	2.05e+02	2.02e+02
f_{33}	Mean	7.64e+02	1.00e+03	5.39e+02	1.28e+03	9.72e+02	9.92e+02	8.15e+02	7.59e+02
f_{34}	Mean	4.02e+02	4.01e+02	4.78e+03	4.34e+03	4.37e+02	3.40e+02	3.60e+02	3.00e+02

arithmetical crossover utilizes the particle's different historical information to construct better offspring. Differential evolution crossover can influence information into the offspring to enhance global exploration. These mechanisms can generate effective guiding vectors that are well diversified and highly qualified. These guiding vectors can better guide the flying of the particles. In order to have good capabilities for exploration and exploitation, a dynamic adjustment strategy is used in PSOCO. In order to verify the performance of PSOCO, lots of benchmarks with different characteristics are adopted in the experiments. Finally, experimental results also show that PSOCO is a competitive PSO variant for some benchmark functions. In a future work, we intend to incorporate different crossover operations into PSOCO. This may result in very competitive algorithms. Obviously, many adaptive methods for DE's crossover control parameter have been proposed. In order to improve the performance of the proposed approach, the design of self-adaptive control parameter strategy is a promising research direction. Finally, we will apply the proposed approach to solve some practical engineering optimization problems.

Acknowledgments

The authors thank the Editor and all the anonymous referees for their constructive comments and valuable suggestions, which are helpful to improve the quality of this paper. This paper is supported by the National Key Research and Development Program of China (Grant nos. 2016YFB0800602 and 2016YFB0800604), the Beijing City Board of Education Science and Technology Key Project (Grant no. KZ201510015015), the Technological Innovation Team of Henan University of Science and Technology (Grant no. 2015XTD011), the Key Technologies Research and Development Program of Henan Province (Grant no. 162102210047) and the Cooperative Engagement Fund of Henan University of Science and Technology (Grant no. 2015ZDCXY03).

References

Ardizzon, G., Cavazzini, G., Pavesi, G., 2015. Adaptive acceleration coefficients for a new search diversification strategy in particle swarm optimization algorithms. *Inform. Sci.* 299, 337–378.

Cai, J., Li, Q., Li, L., Peng, H., Yang, Y., 2012. A hybrid fcaso-sqp method for solving the economic dispatch problems with valve-point effects. *Energy* 38 (1), 346–353.

Chang, W.-D., 2009. Pid control for chaotic synchronization using particle swarm optimization. *Chaos Solitons Fractals* 39 (2), 910–917.

Chang, W.-D., Shih, S.-P., 2010. Pid controller design of nonlinear systems using an improved particle swarm optimization approach. *Commun. Nonlinear Sci. Numer. Simul.* 15 (11), 3632–3639.

Chen, C.-C., 2011. Two-layer particle swarm optimization for unconstrained optimization problems. *Appl. Soft Comput.* 11 (1), 295–304.

Chen, S., 2012. Particle swarm optimization with pbest crossover. In: 2012 IEEE Congress on Evolutionary Computation. IEEE, pp. 1–6.

Chen, W.-N., Zhang, J., Lin, Y., Chen, N., Zhan, Z.-H., Chung, H.S.-H., Li, Y., Shi, Y.-H., 2013. Particle swarm optimization with an aging leader and challengers. *IEEE Trans. Evol. Comput.* 17 (2), 241–258.

Cheng, R., Jin, Y., 2015. A social learning particle swarm optimization algorithm for scalable optimization. *Inform. Sci.* 291, 43–60.

Duong, S.C., Kinjo, H., Uezato, E., Yamamoto, T., 2010. Particle swarm optimization with genetic recombination: a hybrid evolutionary algorithm. *Artif. Life Robot.* 15 (4), 444–449.

Eberhart, R.C., Kennedy, J., 1995. A new optimizer using particle swarm theory. In: Proceedings of the Sixth International Symposium on Micro Machine and Human Science, Vol. 1, New York, NY, pp. 39–43.

Engelbrecht, A.P., 2016. Particle swarm optimization with crossover: a review and empirical analysis. *Artif. Intell. Rev.* 45 (2), 131–165.

García-Martínez, C., Lozano, M., Herrera, F., Molina, D., Sánchez, A.M., 2008. Global and local real-coded genetic algorithms based on parent-centric crossover operators. *European J. Oper. Res.* 185 (3), 1088–1113.

Gaxiola, F., Melin, P., Valdez, F., Castro, J.R., Castillo, O., 2016. Optimization of type-2 fuzzy weights in backpropagation learning for neural networks using gas and pso. *Appl. Soft Comput.* 38, 860–871.

Gong, Y.-J., Li, J.-J., Zhou, Y., Li, Y., Chung, H.S.-H., Shi, Y.-H., Zhang, J., 2016. Genetic learning particle swarm optimization. *IEEE Trans. Cybern.* 46 (10), 2277–2290.

Gülcü, Ş., Kodaz, H., 2015. A novel parallel multi-swarm algorithm based on comprehensive learning particle swarm optimization. *Eng. Appl. Artif. Intell.* 45, 33–45.

Guo, W., Chen, G., Feng, X., 2006. A new strategy of acceleration coefficients for particle swarm optimization. In: 10th International Conference on Computer Supported Cooperative Work in Design, 2006. CSCWD'06, IEEE, pp. 1–5.

Haddar, B., Khemakhem, M., Hanafi, S., Wilbaut, C., 2016. A hybrid quantum particle swarm optimization for the multidimensional knapsack problem. *Eng. Appl. Artif. Intell.* 55, 1–13.

Hansen, N., Ostermeier, A., 2001. Completely derandomized self-adaptation in evolution strategies. *Evol. Comput.* 9 (2), 159–195.

Hao, Z.-F., Wang, Z.-G., Huang, H., 2007. A particle swarm optimization algorithm with crossover operator. In: 2007 International Conference on Machine Learning and Cybernetics, Vol. 2. IEEE, pp. 1036–1040.

Jarboui, B., Damak, N., Siarry, P., Rebai, A., 2008. A combinatorial particle swarm optimization for solving multi-mode resource-constrained project scheduling problems. *Appl. Math. Comput.* 195 (1), 299–308.

Kennedy, J., Eberhart, R., 1995. Particle swarm optimization. In: Proceedings of IEEE International Conference on Neural Networks, Vol. 4, pp. 1942–1948.

Kennedy, J., 1999. Small worlds and mega-minds: effects of neighborhood topology on particle swarm performance. In: Proceedings of the 1999 Congress on Evolutionary Computation, 1999. Vol. 3. CEC 99, IEEE.

Li, Z., Wang, W., Yan, Y., Li, Z., 2015a. Ps-abc: A hybrid algorithm based on particle swarm and artificial bee colony for high-dimensional optimization problems. *Expert Syst. Appl.* 42 (22), 8881–8895.

Li, L., Yang, Y., Peng, H., Wang, X., 2006. Parameters identification of chaotic systems via chaotic ant swarm. *Chaos Solitons Fractals* 28 (5), 1204–1211.

Li, Y., Zhan, Z.-H., Lin, S., Zhang, J., Luo, X., 2015b. Competitive and cooperative particle swarm optimization with information sharing mechanism for global optimization problems. *Inform. Sci.* 293, 370–382.

Li, M., Zhang, Y., Jiang, W., Xie, J., 2009. A particle swarm optimization algorithm with crossover for resource constrained project scheduling problem. In: IITA International Conference on Services Science, Management and Engineering, 2009. SSME'09, IEEE, pp. 69–72.

Li, J., Zheng, S., Tan, Y., 2014. Adaptive fireworks algorithm. In: 2014 IEEE Congress on Evolutionary Computation. (CEC), IEEE, pp. 3214–3221.

Li, J., Zheng, S., Tan, Y., 2017. The effect of information utilization: Introducing a novel guiding spark in the fireworks algorithm. *IEEE Trans. Evol. Comput.* 21 (1), 153–166.

Liang, J., Guo, L., Liu, R., Qu, B., 2015. A self-adaptive dynamic particle swarm optimizer. In: 2015 IEEE Congress on Evolutionary Computation. (CEC), IEEE, pp. 3206–3213.

Liang, J.J., Qin, A.K., Suganthan, P.N., Baskar, S., 2006. Comprehensive learning particle swarm optimizer for global optimization of multimodal functions. *IEEE Trans. Evol. Comput.* 10 (3), 281–295.

Liang, J., Qu, B., Suganthan, P., Hernández-Díaz, A.G., 0000. Problem definitions and evaluation criteria for the cec 2013 special session on real-parameter optimization, Computational Intelligence Laboratory, Zhengzhou University, Zhengzhou, China and Nanyang Technological University, Singapore, Technical Report 201212.

Liang, J., Suganthan, P.N., 2005. Dynamic multi-swarm particle swarm optimizer. In: Swarm Intelligence Symposium, 2005. SIS 2005. Proceedings 2005 IEEE. IEEE, pp. 124–129.

Lin, M., Chin, K., Tsui, K., Wong, T., 2016. Genetic based discrete particle swarm optimization for elderly day care center timetabling. *Comput. Oper. Res.* 65, 125–138.

- Liu, Y., Niu, B., Luo, Y., 2015. Hybrid learning particle swarm optimizer with genetic disturbance. *Neurocomputing* 151, 1237–1247.
- Lu, Y., Liang, M., Ye, Z., Cao, L., 2015. Improved particle swarm optimization algorithm and its application in text feature selection. *Appl. Soft Comput.* 35, 629–636.
- Mendes, R., Kennedy, J., Neves, J., 2004. The fully informed particle swarm: simpler, maybe better. *IEEE Trans. Evol. Comput.* 8 (3), 204–210.
- Miao, Y., Cui, Z., Zeng, J., 2009. Dynamic population-based particle swarm optimization combined with crossover operator. In: *Ninth International Conference on Hybrid Intelligent Systems*, 2009. Vol. 1. HIS'09, IEEE, pp. 399–404.
- Olivas, F., Valdez, F., Castillo, O., Melin, P., 2016. Dynamic parameter adaptation in particle swarm optimization using interval type-2 fuzzy logic. *Soft Comput.* 20 (3), 1057–1070.
- Pan, Q.-K., Tasgetiren, M.F., Liang, Y.-C., 2008. A discrete particle swarm optimization algorithm for the no-wait flowshop scheduling problem. *Comput. Oper. Res.* 35 (9), 2807–2839.
- Pant, M., Thangaraj, R., Abraham, A., 2007. A new pso algorithm with crossover operator for global optimization problems. In: *Innovations in Hybrid Intelligent Systems*. Springer, pp. 215–222.
- Park, J.-B., Jeong, Y.-W., Shin, J.-R., Lee, K.Y., Kim, J.-H., 2007. A hybrid particle swarm optimization employing crossover operation for economic dispatch problems with valve-point effects. In: *International Conference on Intelligent Systems Applications to Power Systems*, 2007. ISAP 2007, IEEE, pp. 1–6.
- Peng, H., Li, L., Kurths, J., Li, S., Yang, Y., 2013. Topology identification of complex network via chaotic ant swarm algorithm. *Math. Probl. Eng.*
- Peram, T., Veeramachaneni, K., Mohan, C.K., 2003. Fitness-distance-ratio based particle swarm optimization. In: *Swarm Intelligence Symposium*, 2003. SIS'03. Proceedings of the 2003 IEEE. IEEE, pp. 174–181.
- Ratnaweera, A., Halgamuge, S.K., Watson, H.C., 2004. Self-organizing hierarchical particle swarm optimizer with time-varying acceleration coefficients. *IEEE Trans. Evol. Comput.* 8 (3), 240–255.
- Samma, H., Lim, C.P., Saleh, J.M., 2016. A new reinforcement learning-based memetic particle swarm optimizer. *Appl. Soft Comput.* 43, 276–297.
- Shen, M., Zhan, Z.-H., Chen, W.-N., Gong, Y.-J., Zhang, J., Li, Y., 2014. Bi-velocity discrete particle swarm optimization and its application to multicast routing problem in communication networks. *IEEE Trans. Ind. Electron.* 61 (12), 7141–7151.
- Shi, Y., Eberhart, R., 1998. A modified particle swarm optimizer. In: *The 1998 IEEE International Conference on Evolutionary Computation Proceedings*, 1998. IEEE World Congress on Computational Intelligence. IEEE, pp. 69–73.
- Shi, Y., Eberhart, R.C., 2001. Fuzzy adaptive particle swarm optimization. In: *Proceedings of the 2001 Congress on Evolutionary Computation*, 2001. Vol. 1. IEEE, pp. 101–106.
- Simon, D., 2008. Biogeography-based optimization. *IEEE Trans. Evol. Comput.* 12 (6), 702–713.
- Singh, D., Singh, V., Ansari, U., 2011. Binary particle swarm optimization with crossover operation for discrete optimization. *Int. J. Comput. Appl.* 28 (11), 15–20.
- Storn, R., Price, K., 1997. Differential evolution—a simple and efficient heuristic for global optimization over continuous spaces. *J. Global Optim.* 11 (4), 341–359.
- Taherkhani, M., Safabakhsh, R., 2016. A novel stability-based adaptive inertia weight for particle swarm optimization. *Appl. Soft Comput.* 38, 281–295.
- Tang, R.-L., Fang, Y.-J., 2015. Modification of particle swarm optimization with human simulated property. *Neurocomputing* 153, 319–331.
- Tawhid, M.A., Ali, A.F., 2016. Simplex particle swarm optimization with arithmetical crossover for solving global optimization problems. *Opsearch* 1–36.
- Valdez, F., Vazquez, J.C., Melin, P., Castillo, O., 2017. Comparative study of the use of fuzzy logic in improving particle swarm optimization variants for mathematical functions using co-evolution. *Appl. Soft Comput.* 52, 1070–1083.
- Wang, S., Phillips, P., Yang, J., Sun, P., Zhang, Y., 2016. Magnetic resonance brain classification by a novel binary particle swarm optimization with mutation and time-varying acceleration coefficients. *Biomed. Eng./Biomed. Tech.* 61 (4), 431–441.
- Wang, H., Wu, Z., Liu, Y., Zeng, S., 2008. Particle swarm optimization with a novel multi-parent crossover operator. In: *Fourth International Conference on Natural Computation*, 2008. Vol. 7. ICNC'08, IEEE, pp. 664–668.
- Wang, L., Yang, B., Chen, Y., 2014. Improving particle swarm optimization using multi-layer searching strategy. *Inform. Sci.* 274, 70–94.
- Wu, G., Qiu, D., Yu, Y., Pedrycz, W., Ma, M., Li, H., 2014. Superior solution guided particle swarm optimization combined with local search techniques. *Expert Syst. Appl.* 41 (16), 7536–7548.
- Xie, J., Yang, J., 2010. A novel crossover operator for particle swarm algorithm. In: *2010 International Conference on Machine Vision and Human-Machine Interface*. IEEE, pp. 161–164.
- Xu, X., Tang, Y., Li, J., Hua, C., Guan, X., 2015. Dynamic multi-swarm particle swarm optimizer with cooperative learning strategy. *Appl. Soft Comput.* 29, 169–183.
- Yao, X., Liu, Y., Lin, G., 1999. Evolutionary programming made faster. *IEEE Trans. Evol. Comput.* 3 (2), 82–102.
- Yao, Y., Ni, Q., Lv, Q., Huang, K., 2015. A novel heterogeneous feature ant colony optimization and its application on robot path planning. In: *2015 IEEE Congress on Evolutionary Computation*. (CEC), IEEE, pp. 522–528.
- Zambrano-Bigiarini, M., Clerc, M., Rojas, R., 2013. Standard particle swarm optimisation 2011 at cec-2013: A baseline for future pso improvements. In: *2013 IEEE Congress on Evolutionary Computation*. IEEE, pp. 2337–2344.
- Zhan, Z.-H., Zhang, J., Li, Y., Shi, Y.-H., 2011. Orthogonal learning particle swarm optimization. *IEEE Trans. Evol. Comput.* 15 (6), 832–847.
- Zhang, T., Hu, T., Guo, X., Chen, Z., Zheng, Y., 2013. Solving high dimensional bilevel multiobjective programming problem using a hybrid particle swarm optimization algorithm with crossover operator. *Knowl.-Based Syst.* 53, 13–19.
- Zhang, C., Ouyang, D., Ning, J., 2010. An artificial bee colony approach for clustering. *Expert Syst. Appl.* 37 (7), 4761–4767.
- Zhang, Y., Wang, S., Ji, G., 2015. A comprehensive survey on particle swarm optimization algorithm and its applications. *Math. Probl. Eng.*
- Zhang, Y., Wang, S., Phillips, P., Ji, G., 2014. Binary pso with mutation operator for feature selection using decision tree applied to spam detection. *Knowl.-Based Syst.* 64, 22–31.
- Zhao, X., Lin, W., Hao, J., Zuo, X., Yuan, J., 2016. Clustering and pattern search for enhancing particle swarm optimization with euclidean spatial neighborhood search. *Neurocomputing* 171, 966–981.
- Zheng, S., Janecek, A., Li, J., Tan, Y., 2014. Dynamic search in fireworks algorithm. In: *2014 IEEE Congress on Evolutionary Computation*. (CEC), IEEE, pp. 3222–3229.
- Zheng, S., Janecek, A., Tan, Y., 2013. Enhanced fireworks algorithm. In: *2013 IEEE Congress on Evolutionary Computation*. IEEE, pp. 2069–2077.